

Unit 1, Lesson 3: Rational Numbers and Absolute Value

Benchmarks

6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.

6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite directions using rational numbers. Compare them on a number line and explain the meaning of zero within its context.

6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.

Learning Targets

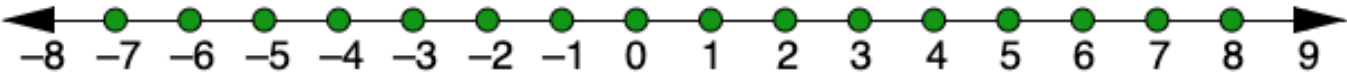
- ☐ I can plot, compare, and order rational numbers in a mathematical or real-world context.
- ☐ I can compare and perform operations involving the absolute value of two numbers.

1.3.1 Warm-Up: Math Talk

1. Are there more inches in a mile or seconds in a day?
2. Discuss with your partner and justify your reasoning mathematically.

1.3.2 Refresher: Rational Numbers, Integers, and Negative Numbers

The values in green on the number line are some of the values in the set of *integers*.



The set of *rational numbers* includes all integers as well as the fractional parts between integers on the number line.



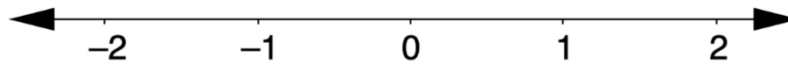
Rational numbers are numbers that can be expressed as the ratio of two integers.

- 1. Use the definition of rational numbers to write each number as a ratio of two integers.

Number	Ratio of Two Integers
-1.5	
9	

1.3.3 Guided Practice: Plotting Rational Numbers on the Number Line

1. Use the number line shown to complete the following.

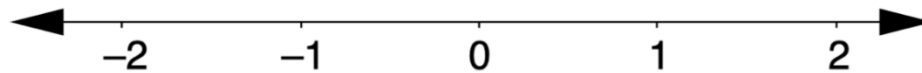


- Make a tick mark halfway between each value marked on the number line. Label each tick mark with the value of its location.
- Make another tick mark halfway between each value and the tick marks that were previously made. Label each with the value of its location.
- Plot and label the values $-1\frac{7}{12}$ and 0.25 on the number line.

2. Four values are shown.

0.4 - 15% 12% - 0.25

- Plot each of the values on the number line. Label each point with its numeric value.



- Complete the inequality statement to show the relationship between the values.

_____ < _____ < _____ < _____

3. Complete the statements.

- When moving from left to right ($\rightarrow$) on the number

line, the values

- ☐ increase.

☐ decrease.

This can be

described as moving in a

- ☐ positive

☐ negative

direction.

b. When moving from right to left ($\leftarrow$) on the number

line, the values

- ☐ increase.
- ☐ decrease

This can be

described as moving in a

- ☐ positive
- ☐ negative

direction.

4. Order these values from least to greatest.

0.43, -4.2, 0, 50%, -1.5, -6.712, 19%

_____, _____, _____, _____, _____, _____, _____

1.3.4 Guided Practice: Absolute Value

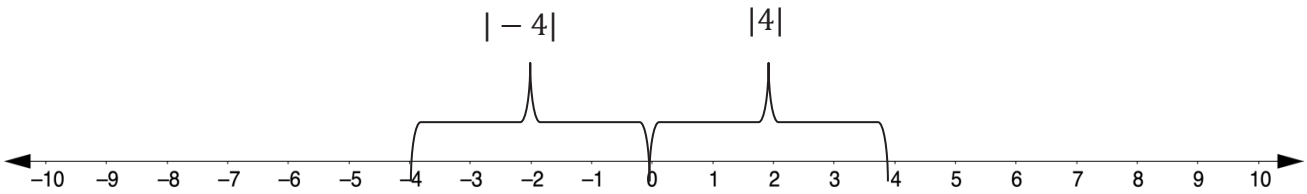
1. Using words from the bank below, complete the statements. Not all words will be used.

zerointegerdistanceabsolute value

Pairs of opposite values are the same _____ from zero on the number line. They are mirror images of each other, with a line of symmetry through _____. The distance a number is from zero on a number line is called _____.

2. Complete the statement.

The absolute value of 4 and -4 is _____ because they are both 4 units from 0 on the number line.



The symbol for absolute value is two vertical bars around a number. The absolute value of 4 is written as $|4|$.

$|4| = 4$ and $|-4| = 4$

3. Compare each pair of values and their absolute values.
- a. For each row, use the symbols $>$, $<$, or $=$ to complete each inequality.

	Inequality	Absolute Value Inequality
Pair 1	$4 \underline{\hspace{1cm}} -5$	$ 4 \underline{\hspace{1cm}} -5 $
Pair 2	$-3 \underline{\hspace{1cm}} 2$	$ -3 \underline{\hspace{1cm}} 2 $
Pair 3	$6 \underline{\hspace{1cm}} -6$	$ 6 \underline{\hspace{1cm}} -6 $

- b. Discuss what you notice about the comparisons with your partner. Then, summarize your discussion.

4. Complete the statement.

When comparing the absolute values of two numbers,
numbers that are closer to zero will always have the

- ☐ larger
 - ☐ smaller

absolute value.

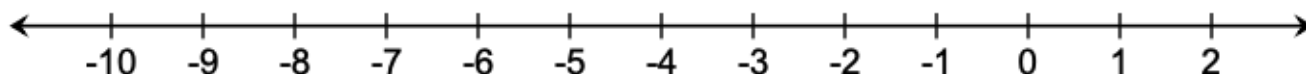
5. Dariel owes his friend \$15.50.

- a. Sketch a number line that represents this situation.
- b. Represent this situation as a rational number.
- c. What is the absolute value of the rational number?
- d. What does 0 represent in this situation?

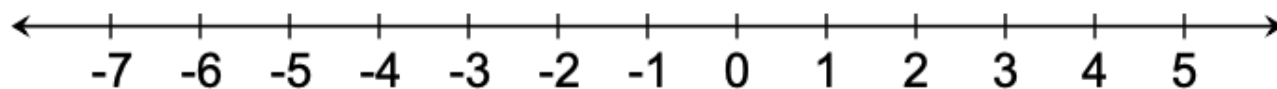
1.3.5 Try It: Absolute Value

1. Complete each statement and model it on the number line given.

a. $|-7.8| = \underline{\hspace{2cm}}$



b. $|\underline{\hspace{1cm}}| = 3\frac{1}{2}$



2. What must be true about two numbers that have the same absolute value?

3. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

$$|-42| \underline{\hspace{1cm}} -25$$

$$-21 \underline{\hspace{1cm}} |18|$$

$$|-582| \underline{\hspace{1cm}} |-790|$$

1.3.6 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like to have answered...	
One mistake I learned from today...	One thing I am not sure about...YET!
One thing I am confident I know is...	

